



جامعة مصر للمعلوماتية
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Computer and Information Systems
Data Analysis Course

Social Media and Its Side Effects

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Introduction

In today's digital age, social media has become an inseparable part of daily life, particularly among young people. Platforms such as Instagram, TikTok, and WhatsApp are used for hours each day, raising important questions about their impact on mental well-being, focus, and self-perception. While social media offers many benefits such as communication, entertainment, and access to information, growing concerns suggest it may also carry significant side effects. This study investigates whether spending more time on social media negatively affects users' focus, emotional state, and self-image, with the aim of providing a data-driven understanding of these effects among young users.

Research Question

Does spending more time on social media have a negative effect on users' focus, emotional state, and self-image?

Hypothesis

It is hypothesized that users who spend more time on social media will report lower focus levels, more mood changes triggered by likes, and a greater tendency to compare themselves to others, compared to users who spend less time on social media.

Population of Interest:

The population of interest for this study is young social media users, primarily students and young adults, who regularly use platforms such as Instagram, TikTok, WhatsApp, and Snapchat. This age group was chosen because they are among the most active users, and their digital habits can noticeably impact their focus, emotions, and self-perception during a critical stage of life.

Sampling Method:

This study employed a convenience sampling approach, where the survey was created using Google Forms and distributed digitally through WhatsApp groups, including both university and non-university groups, as well as directly to classmates and friends from different social circles. This method was chosen because it is practical for student research with limited time and resources, and because our personal networks are largely composed of the target demographic of young social media users. While convenience sampling is not fully random, the survey was shared across multiple channels rather than a single social circle to reach a broader and more varied sample. A total of 91 valid responses were collected, providing a sufficient sample size for meaningful descriptive statistical analysis.

Bias Identification:

In designing this survey, several potential sources of bias were identified and addressed:

Sampling Bias: The survey was distributed through personal and university networks, meaning respondents tend to skew toward young adults who are already active on social media. To reduce this, the survey was shared across multiple channels rather than a single group.

Social Desirability Bias: Some respondents may have underreported their screen time or negative effects to avoid appearing dependent on social media. To minimize this, all responses were kept anonymous and questions were worded in a neutral, non-judgmental way.

Question Neutrality: A 'Maybe' option was included in relevant questions to allow honest responses without forcing a yes or no answer.

Recall Bias: Screen time was measured using bracketed ranges rather than exact figures to make responses more reliable and reduce the burden of memory on participants

Survey Questions:

- What is your current age? (Under 20 / 20+)
- What is your gender?
- Which social media platforms do you frequently use?
- What is the type of content you usually see on social media?
- What is your average daily screen time? (0–3 hrs / 3–7 hrs / 7+ hrs)
- Did you ever regret using social media at any point in your day?
- Why do you open social media most of the time?
- Did the number of likes ever affect your mood?
- On a scale of 1–10, how much does social media negatively affect your focus?
- On a scale of 1–10, how good is your focus when not using social media?
- Did you ever compare yourself with anyone on social media?
- On a scale of 1–10, how negatively does social comparison affect your mood?

Online survey link:

https://docs.google.com/forms/d/e/1FAIpQLSdnR160X_i2nCFzdRER6LE6Z1Wb957I_sxIN-HOY79X6IbjbA/viewform?usp=publish-editor

Number of samples collected: 91

Analysis:

Descriptive Statistics:

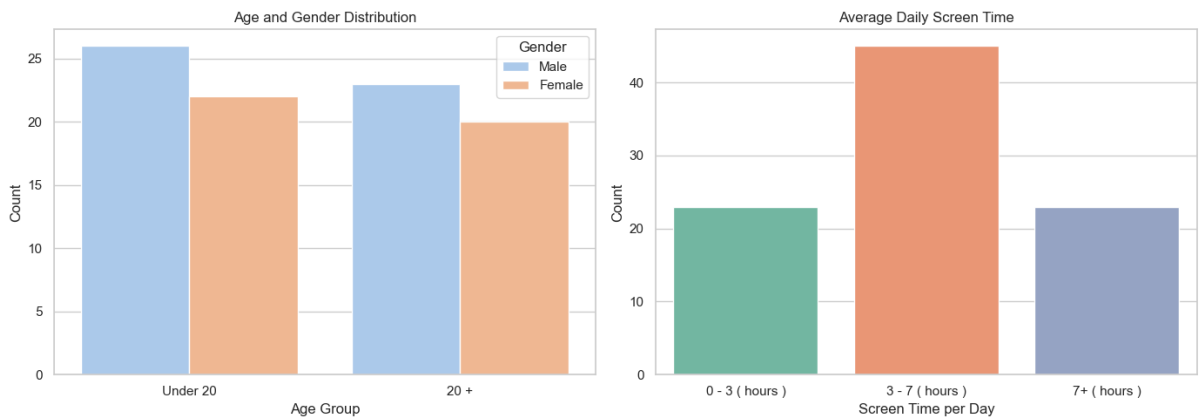
We analyzed the data by calculating the mean, median, and mode for the three scale-based questions. We also created charts to identify trends and patterns across the 91 responses.

The table below summarizes the key findings:

Metric	Mean	Median	Mode
Social media bad effect on focus	6.68	7	6
Focus quality without social media	7.54	8	8
Negative mood impact from self-comparison	4.30	4	1

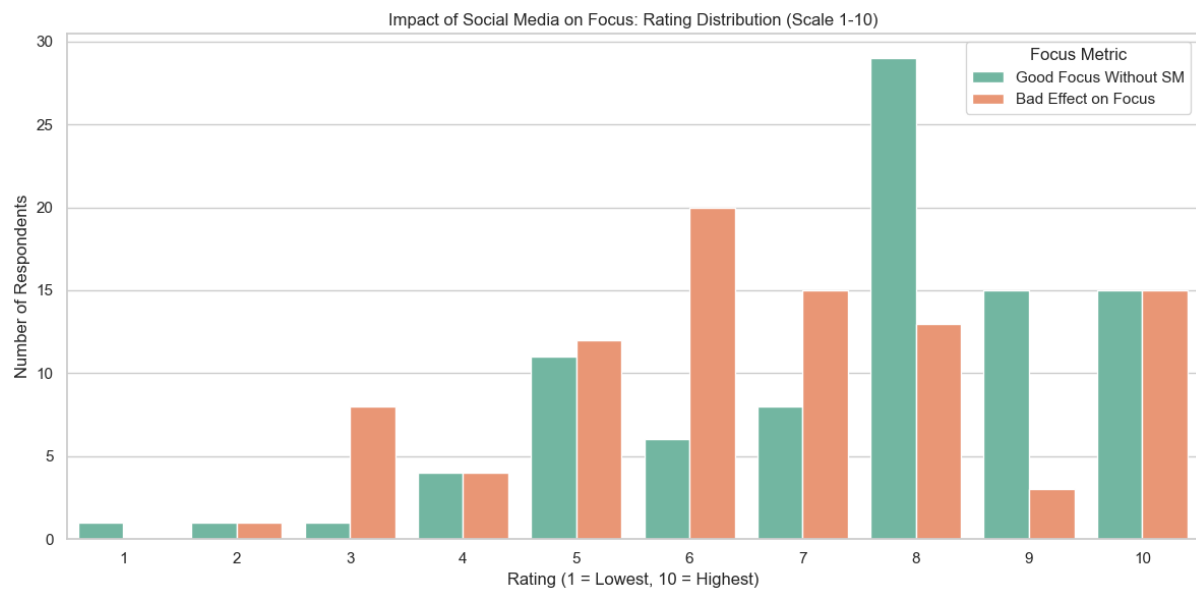
Visual Analysis:

Figure 1: Age, Gender, and Daily Screen Time Distribution



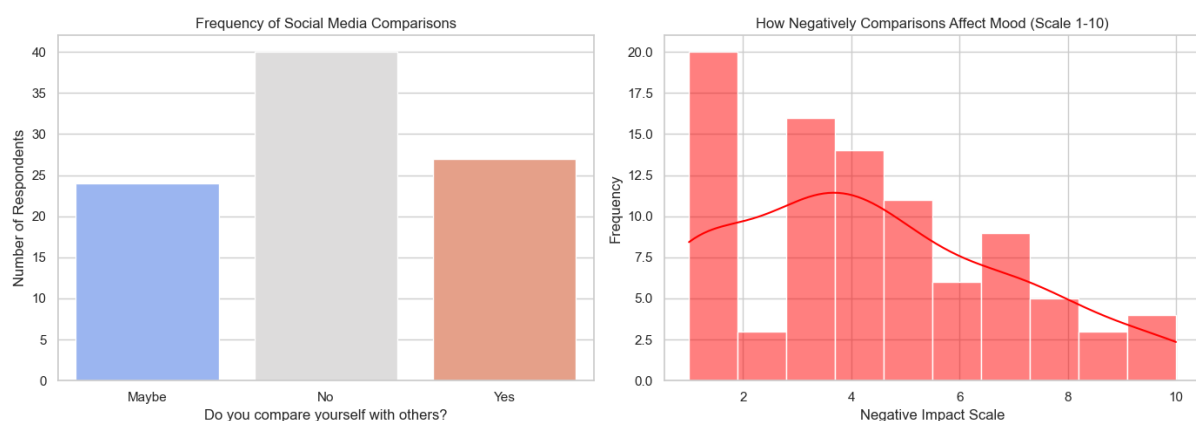
The chart reveals that the sample is nearly evenly split between respondents under 20 and those aged 20 and above, with a slight male majority. The screen time distribution shows that most respondents fall into the 3 to 7 hours per day bracket, confirming that heavy social media engagement is the norm rather than the exception among our sample. Additionally, female respondents reported higher daily screen time than males, with more females falling into the 7+ hours category.

Figure 2: Impact of Social Media on Focus (Scale 1-10)



This chart compares two focus metrics on the same 1 to 10 scale: how badly social media affects focus, and how good focus is without it. The bad-effect scores cluster heavily between 5 and 10, with 6 being the most frequent rating at 20 responses, followed by 7 and 10 with 15 each, indicating that the large majority of respondents feel social media meaningfully disrupts their concentration. In contrast, focus without social media skewed even higher, with 8 being the most common rating at 29 responses, followed by 9 and 10 with 15 each, confirming that respondents consistently feel more focused when away from social media.

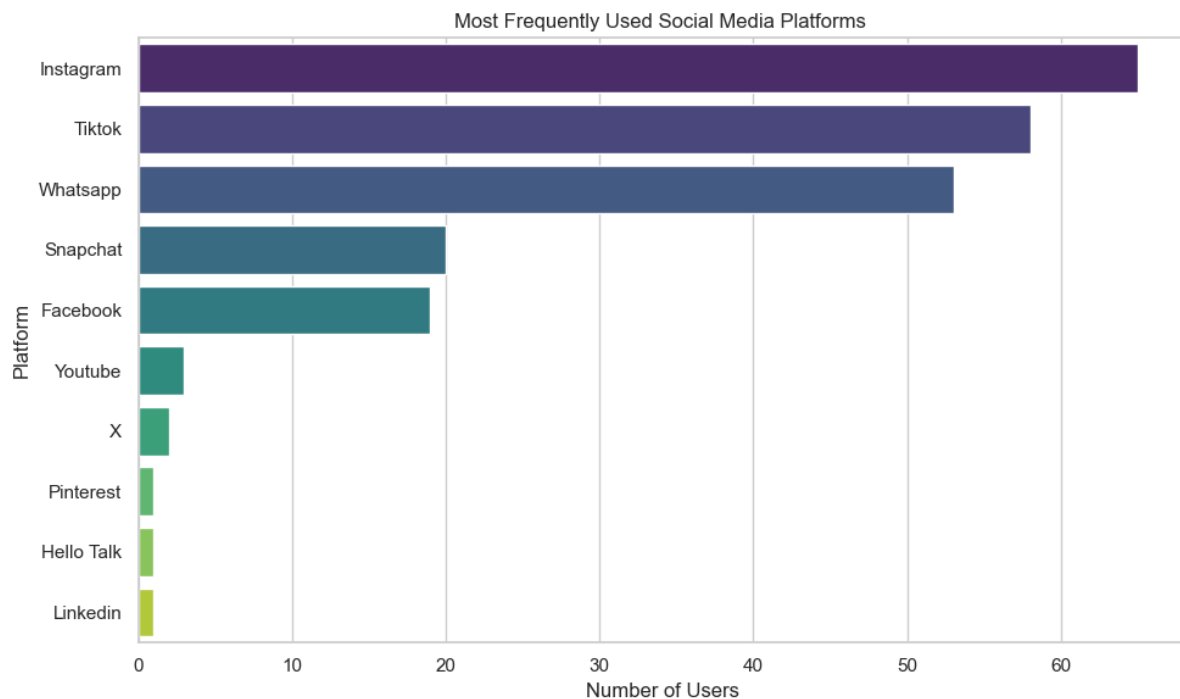
Figure 3: Frequency of Self-Comparison and Its Effect on Mood



The chart shows that 40 respondents said they do not compare themselves to others on social media, while 27 said yes and 24 were unsure, meaning that more than half acknowledge some degree of comparison behaviour. The mood impact scores are skewed toward the lower end of the scale, with a mode of 1 and a mean of 4.30, suggesting that

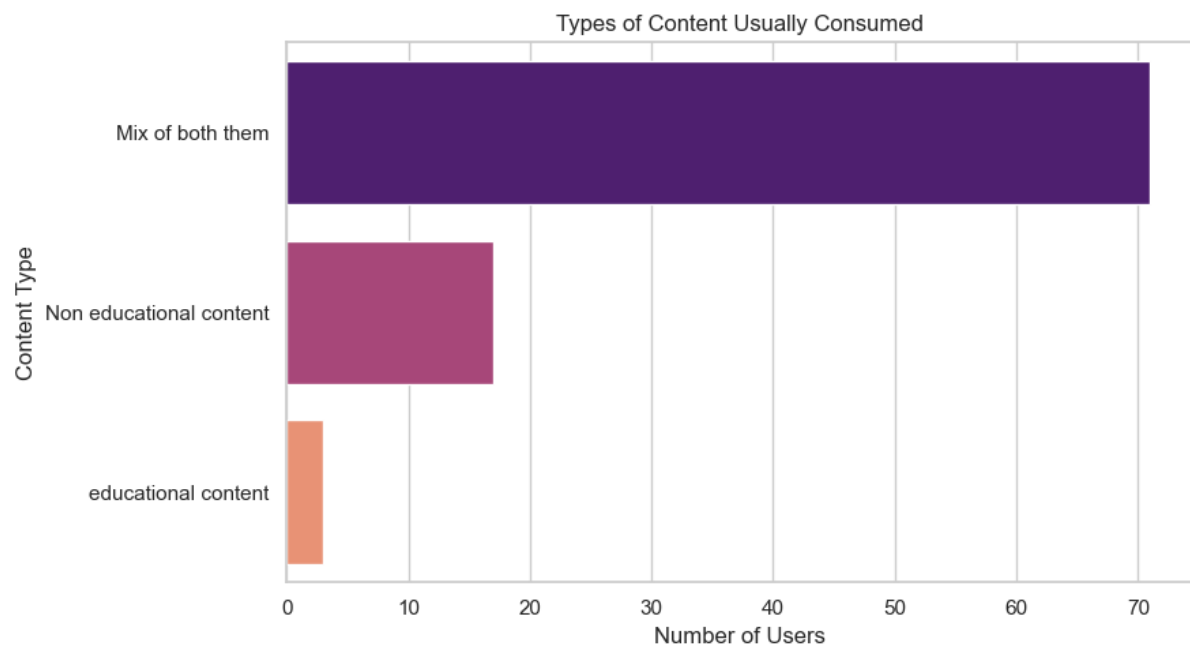
while comparison is common, its emotional effect is mild for most users. However, a notable portion rated the negative impact at 7 or above, indicating that for some users comparison can have a more serious emotional consequence.

Figure 4: Most Frequently Used Social Media Platforms



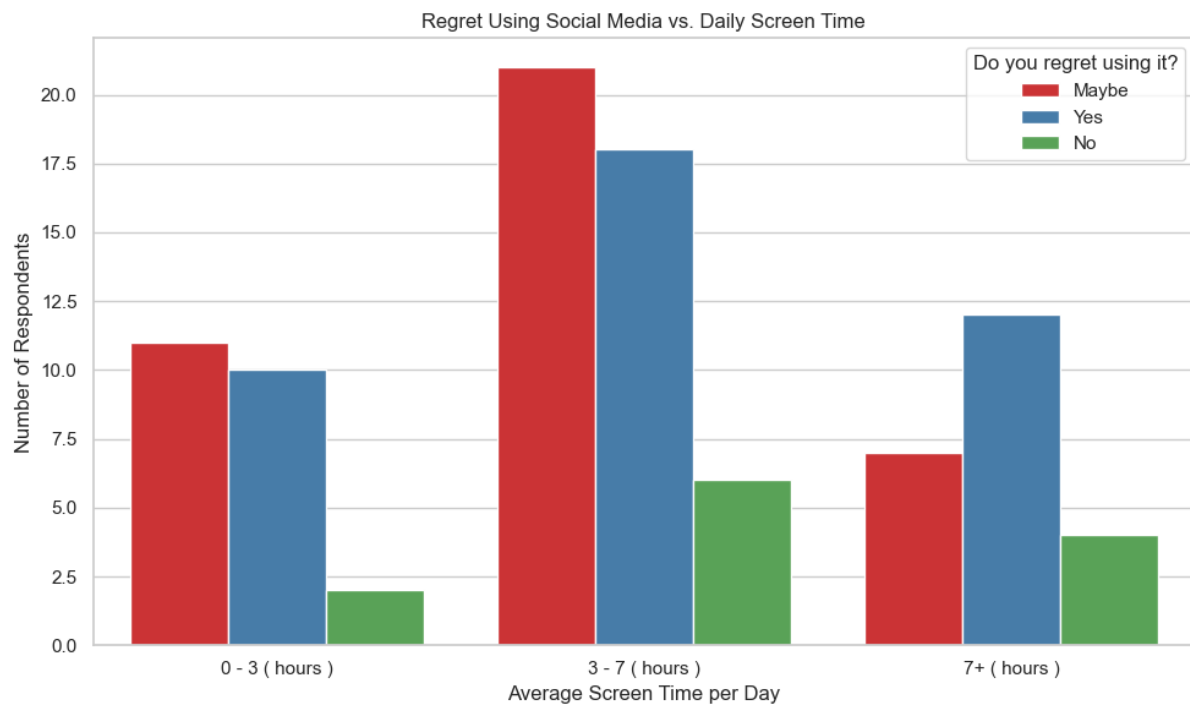
Instagram is the most widely used platform among respondents, with 65 users, followed by TikTok with 58 users and WhatsApp with 53. Since Instagram and TikTok are primarily visual, scroll-based platforms designed to maximize engagement time, their dominance helps explain why respondents reported significant negative effects on focus and mood shifts driven by self-comparison. Notably, TikTok's strong presence also reflects its rapid growth among Egyptian youth, which aligns with broader regional trends.

Figure 5: Content Type



The chart reveals that mixed content consumption dominates among respondents, with 71 out of 91 reporting that they consume both educational and non-educational content on social media. Non-educational content only came second with 17 respondents, while purely educational consumption was minimal at just 3. This pattern suggests that entertainment remains a central part of social media usage among our sample, and the scarcity of purely educational engagement may contribute to the negative effects on focus and mood observed in this study.

Figure 6: Regret vs. Screen Time



The chart shows that regret is most common among respondents who spend 3 to 7 hours daily on social media, which is the largest group in our sample. Notably, only 12 respondents across all screen time groups reported no regret at all, meaning approximately 87% experienced at least some level of dissatisfaction with their usage. Among heavy users spending 7 or more hours daily, 12 out of 23 answered yes, the highest yes rate across all groups, strongly supporting the idea that increased screen time is linked to greater post-usage regret.

Conclusion

Based on data collected from 91 respondents, the findings largely support the hypothesis that spending more time on social media negatively affects users' focus, emotional state, and self-image. Focus disruption was clearly evident, with respondents rating social media's bad effect on focus at a mean of 6.68 out of 10, while reporting significantly better focus at 7.54 when away from it. Regret was also widespread, with approximately 87% of respondents acknowledging some level of dissatisfaction with their usage, especially among heavy users who spent 7 or more hours daily. However, the negative mood impact from self-comparison scored a mean of only 4.30 out of 10, suggesting that while comparison is fairly common, its emotional effect is less severe than expected. Overall, the data suggests that spending more time on social media is consistently linked to reduced focus and increased regret, though its influence on mood and self-image varies from person to person.

Critical Reflection and Potential Issues

While this study produced meaningful insights, several limitations and potential issues were identified upon reflection:

Convenience Sampling Limits Generalizability: The sample skewed toward younger respondents, predominantly university students, which may limit how well the findings apply to older age groups or non-students who may have different social media habits.

Self-Reported Data is Approximate: Since all responses were self-reported, there is no way to fully verify their accuracy. Respondents estimated their screen time using broad ranges rather than exact figures, and actual usage data from phone screen-time tools would produce more precise findings.

Correlation vs. Causation: The data reveals associations rather than causal relationships. It is possible that individuals already experiencing focus difficulties or low mood are more drawn to social media, rather than social media being the direct cause of these outcomes.

Hidden Bias in Regret Measurement: The high regret rate of 87% may partially reflect a social expectation that excessive social media use is considered bad, leading some respondents to report regret even when their actual discomfort is minimal.

Despite these limitations, this study provides a useful and data-driven snapshot of how young social media users experience its side effects, offering a foundation for further and more rigorous research on this topic.